

NOVA KAKHOVKA, Ukraine

WMO: 338690

Lat: 46.7892N

Lon: 33.3586E

Elev: 25

StdP: 101.03

Time Zone: 2.00 (EUE)

Period: 86-16

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-14.2	-11.8	-17.3	0.8	-13.2	-15.1	1.0	-11.0	10.7	-2.7	9.5	-5.4	3.6	20	0.465

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.2	32.0	20.9	30.6	20.3	29.3	19.8	22.3	29.1	21.5	28.5	20.7	27.4	2.6	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
20.1	14.9	25.1	19.1	14.0	24.5	18.3	13.2	23.7	65.6	29.3	62.7	28.5	59.9	27.3	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	9.4	7.7	6.6	DB	-16.8	34.6	3.0	1.8	-19.0	35.9	-20.7	36.9	-22.4	37.9	-24.7	39.3
(5)				WB	-17.2	24.0	3.0	1.6	-19.3	25.1	-21.1	26.0	-22.8	26.9	-24.9	28.1

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-1.3	-0.3	4.0	10.6	16.1	20.3	23.0	22.8	17.6	11.1	4.3	-0.1
	DBStd	9.77	5.39	4.95	4.81	3.59	3.50	2.98	2.72	3.14	3.94	4.81	4.79	4.48
	HDD10.0	1403	351	288	189	36	2	0	0	0	1	44	179	313
	HDD18.3	3189	609	521	445	231	85	14	2	3	59	227	422	571
	CDD10.0	1679	0	1	3	55	193	309	404	398	228	80	7	1
	CDD18.3	422	0	0	0	1	17	73	148	143	37	3	0	0
	CDH23.3	3524	0	0	0	4	112	523	1296	1332	248	9	0	0
	CDH26.7	1087	0	0	0	1	9	125	416	477	59	1	0	0
Wind	WSAvg	2.9	3.2	3.3	3.4	3.0	2.6	2.4	2.4	2.5	2.5	3.3	3.3	3.4
Precipitation	PrecAvg	418	30	27	29	31	40	53	38	33	38	31	33	32
	PrecMax	629	71	78	59	78	114	101	95	100	116	91	78	92
	PrecMin	256	6	4	4	3	4	21	3	2	4	3	0	4
	PrecStd	91	17	16	14	23	25	21	27	23	29	19	21	21
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.3	13.8	17.7	23.7	27.8	31.6	33.7	33.9	31.3	24.5	16.6	12.2
		MCWB	7.2	8.9	11.2	14.7	18.1	20.4	22.1	21.2	20.8	16.3	13.5	10.2
	2%	DB	7.7	9.8	14.5	20.9	26.2	29.7	31.8	32.2	28.2	22.1	14.2	9.2
		MCWB	6.4	7.5	9.3	13.4	17.2	19.9	21.0	20.8	19.1	15.6	12.3	7.2
	5%	DB	6.3	7.5	12.2	19.1	24.6	28.0	30.4	30.9	26.3	19.7	12.7	7.3
		MCWB	5.1	5.7	7.7	12.1	16.3	19.2	20.5	20.1	18.0	14.7	11.1	5.9
	10%	DB	4.8	5.7	10.1	17.2	22.9	26.5	29.0	29.5	24.0	17.7	10.7	5.7
		MCWB	3.8	4.2	6.4	11.3	15.7	18.9	20.0	19.8	17.3	13.9	9.2	4.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.0	9.5	11.5	15.6	19.4	22.0	23.7	22.8	21.7	18.3	13.9	10.7
		MCDB	8.8	12.2	16.7	22.5	25.2	28.7	29.6	30.3	28.1	21.3	15.5	11.0
	2%	WB	6.6	7.7	9.8	14.0	18.4	20.9	22.8	21.9	20.1	16.9	12.8	7.4
		MCDB	7.4	9.7	13.7	19.5	23.9	27.0	28.8	29.9	26.3	20.1	14.0	8.8
	5%	WB	5.2	5.9	8.3	12.9	17.5	20.2	21.9	21.2	19.0	15.7	11.2	6.1
		MCDB	6.1	7.2	11.3	18.2	22.7	26.3	28.4	29.0	23.7	18.8	12.6	7.2
	10%	WB	3.8	4.3	7.1	11.8	16.6	19.5	21.1	20.4	18.1	14.4	9.4	4.7
		MCDB	4.7	5.6	9.5	16.1	21.7	25.1	27.4	27.7	22.8	17.3	10.5	5.6
Mean Daily Temperature Range	5% DB	MDBR	5.0	5.8	7.0	9.0	9.6	9.3	10.2	10.4	9.4	7.7	5.2	4.8
		MCDBR	6.2	8.5	11.0	12.4	11.6	11.6	11.8	11.9	11.7	10.0	7.2	6.4
		MCWBR	4.9	6.0	6.2	5.9	4.9	4.5	4.5	4.9	4.8	5.2	5.4	4.8
	5% WB	MCDBR	5.7	7.9	10.3	11.4	10.3	10.1	10.6	11.0	10.2	9.1	6.7	6.0
		MCWBR	4.8	6.0	6.3	6.1	4.7	4.5	4.8	5.1	5.2	5.5	5.5	4.7
Clear-Sky Solar Irradiance	taub	0.339	0.368	0.395	0.444	0.419	0.421	0.445	0.442	0.413	0.376	0.357	0.340	
	taud	2.291	2.229	2.203	2.094	2.255	2.279	2.222	2.217	2.263	2.346	2.331	2.313	
	Ebn at Noon	727	781	818	810	848	847	819	804	791	766	700	679	
	Edh at Noon	79	103	122	149	132	130	136	131	114	91	76	70	
All-Sky Solar Radiation	RadAvg	1.04	1.92	3.15	4.81	6.06	6.41	6.42	5.77	4.14	2.55	1.22	0.85	
	RadStd	0.18	0.25	0.33	0.41	0.51	0.43	0.38	0.30	0.43	0.30	0.18	0.14	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	+0.73	N/A	N/A	+1.03	N/A	N/A	-110	-153	+124	+82

Nomenclature: See separate page